Selected Problems (University Physics)

Chapter 1 Kinematics of a Particle

1-4

A particle moves along the x axis with acceleration $a = 4 + 3t$, where t is in s and a in $\mathrm{m \cdot s^{-2}}$. At $t = 0$, $x = 2\ \mathrm{m}$ and $v = 5\ \mathrm{m \cdot s^{-1}}$. Find the velocity and position of the particle at $t = 10\mathrm{s}$.

1-6

A particle moves in a straight line; its instantaneous acceleration varies as $a = -A\omega^2 \cos \omega t$. At $t = 0$, $v_x = 0$ and $x = A$, where A and ω are positive constants. Find the kinematic equation of the particle.

Chapter 2 Dynamics of a Particle

2-7

A particle of mass $1\ \mathrm{kg}$ moves in a straight line under a force $F = 3 + 4t$ (SI), starting from $v_0 = 10\ \mathrm{m \cdot s^{-1}}$ (the direction of v_0 is the same as that of the force). Find:

(1) the impulse I of the force during $0 \sim 2\ \mathrm{s}$;

(2) the velocity of the particle at $t = 2\ \mathrm{s}$.

2-10

While a bullet travels down a gun barrel, the net force on it has magnitude $F = 400 - \frac{4}{3} \times 10^5 t$, and its speed when it leaves the muzzle is $300\ \mathrm{m \cdot s^{-1}}$. The net force happens to be zero exactly where the bullet leaves the barrel. Find:

(1) the time t the bullet takes to travel the full length of the barrel;

(2) the impulse I of the force on the bullet while it is in the barrel;

(3) the mass m of the bullet.

2-16

An object of mass $2\ \mathrm{kg}$ moves in the positive x direction under a variable force $F_x = 4 + 6x$ (SI). At $t = 0$, $x_0 = 0$ and $v_0 = 0$. Find:

(1) the work done by the variable force as the object moves from $x = 0$ to $x = 4\mathrm{m}$;

(2) the speed of the object at $x = 4\mathrm{m}$;

(3) the impulse of the variable force during the motion from $x = 0$ to $x = 4\mathrm{m}$.

2-18

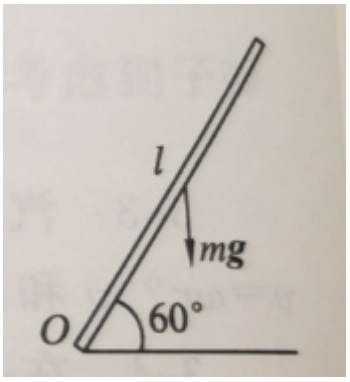
A particle of mass $m = 0.5\ \mathrm{kg}$ moves in a plane with kinematic equations $x = 5t$, $y = 0.5t^2$ (SI). Find the work done by the external force on the particle from $t = 2\mathrm{s}$ to $t = 4\mathrm{s}$.

Chapter 3 Rotation of a Rigid Body about a Fixed Axis

3-6

A uniform straight rod of length $1\mathrm{m}$ can rotate about a smooth, horizontal, fixed axis through one end and perpendicular to the rod. The other end is raised so that the rod makes a 60° angle with the horizontal, and the rod is then released from rest, as shown in the figure. The moment of inertia of the rod about the axis is $\frac{1}{3}ml^2$, where m and l are the mass and length of the rod. Find:

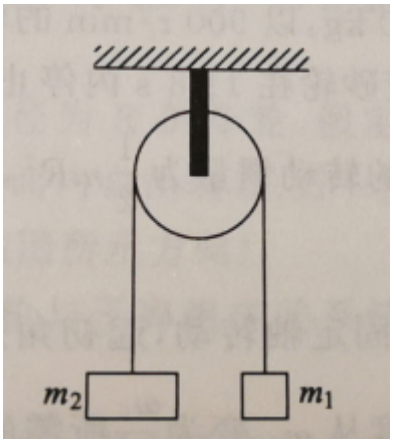
Figure for Exercise 3-6



3-8

An Atwood machine is used to measure the moment of inertia of a pulley. As shown in the figure, the two ends of the string over the pulley carry weights of mass $m_1 = 0.46\ \mathrm{kg}$ and $m_2 = 0.50\ \mathrm{kg}$ respectively; the pulley has radius $0.05\mathrm{m}$, and friction at the axle and the mass of the string are negligible. Starting from rest, the weights are released, and in $5.0\mathrm{s}$ a weight is measured to descend $0.75\mathrm{m}$. What is the moment of inertia of the pulley?

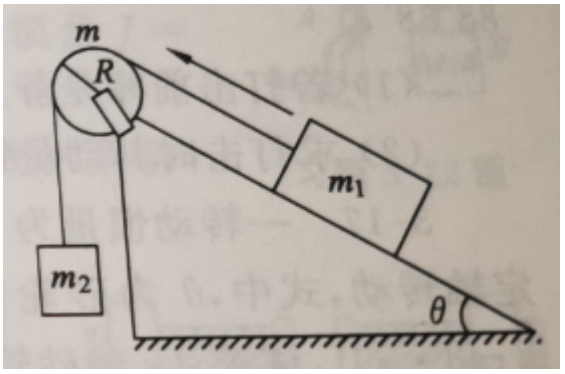
Figure for Exercise 3-8



3-10

As shown in the figure, a block of mass $m_1 = 5\ \mathrm{kg}$ slides down an incline of angle $\theta = 30^\circ$ with coefficient of kinetic friction $\mu = 0.25$. The block is tied to a string that passes over a fixed pulley and carries a weight of mass $m_2 = 10\ \mathrm{kg}$ at its other end. The string does not slip on the pulley; the pulley, a uniform disk of mass $m = 20\ \mathrm{kg}$ and radius $R = 20\ \mathrm{cm}$. Find the acceleration a of the weight and the tension T of the string.

Figure for Exercise 3-10



Chapter 6 Mechanical Oscillations and Mechanical Waves

6-5

A particle undergoes simple harmonic motion with period $2\mathrm{s}$ and amplitude $0.06\mathrm{m}$; at $t = 0$ the particle is exactly at the negative-direction maximum displacement. Find:

(1) the kinematic equation of the particle;

(2) the wave expression of the one-dimensional simple harmonic wave formed when this oscillation propagates in the positive x direction at wave speed $u = 2\ \mathrm{m} \cdot \mathrm{s}^{-1}$ (take the equilibrium position of the particle as the origin);

(3) the wavelength of the wave.

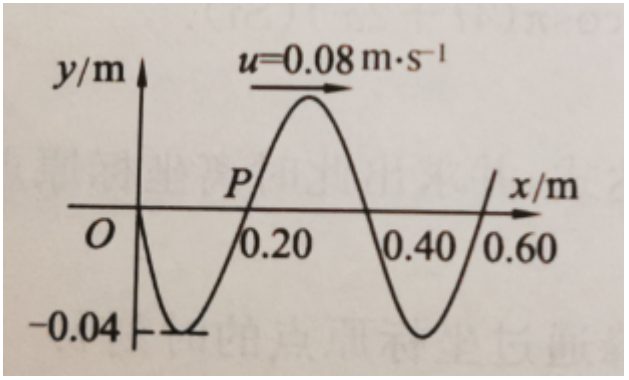
6-10

The figure shows the waveform of a plane simple harmonic wave at $t = 0$. Find:

(1) the wave expression of the wave;

(2) the equation of motion of the particle at P ($x = 0.20\ \mathrm{m}$).

Figure for Exercise 6-10



Chapter 9 Electrostatic Field

9-9

Positive charge of volume density $\rho = 10^{-9}\ \mathrm{C \cdot m^{-3}}$ is uniformly distributed between two hypothetical concentric spherical surfaces of radii $10\mathrm{cm}$ and $20\mathrm{cm}$. Find the electric field strength at $5\mathrm{cm}$, $15\mathrm{cm}$ and $50\mathrm{cm}$ from the center.

9-12

An "infinitely long" straight circular cylinder of radius R is uniformly charged with volume charge density ρ. Find the distribution of its electric field strength and sketch the E-r curve.

9-22

Two concentric spherical surfaces have radii $10\mathrm{cm}$ and $30\mathrm{cm}$. The smaller surface carries positive charge $10^{-8}\ \mathrm{C}$ uniformly distributed, and the larger one positive charge $1.5 \times 10^{-8}\ \mathrm{C}$. Find the electric potential at $20\mathrm{cm}$ and $50\mathrm{cm}$ from the center.

Chapter 10 Steady Magnetic Field

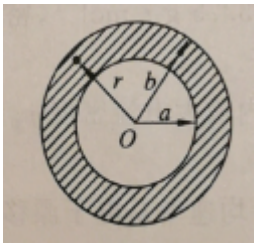
10-9

A hollow cylindrical conductor has inner and outer radii a and b and carries a current I. The current I is uniformly distributed over the cross-section of the conductor. Prove that the magnitude of the magnetic induction B at every point inside the conductor ($a < r < b$) is given by

$$B = \frac{\mu_0 I}{2\pi(b^2 - a^2)} \frac{r^2 - a^2}{r}$$

Check this formula in the limiting case $a = 0$. What is it for $r \geq b$?

Figure for Exercise 10-9



10-10

A very long coaxial cable consists of a solid cylindrical conductor (radius a) and a coaxial cylindrical shell (inner and outer radii b, c), as shown. In use, a current I flows out through one conductor and returns through the other, both uniformly distributed over their cross-sections. Find the magnitude of the magnetic induction in each region:

(1) inside the inner conductor ($r < a$)

(2) between the two conductors ($a < r < b$)

(3) within the shell ($b < r < c$)

(4) outside the cable ($r > c$)

Figure for Exercise 10-10

